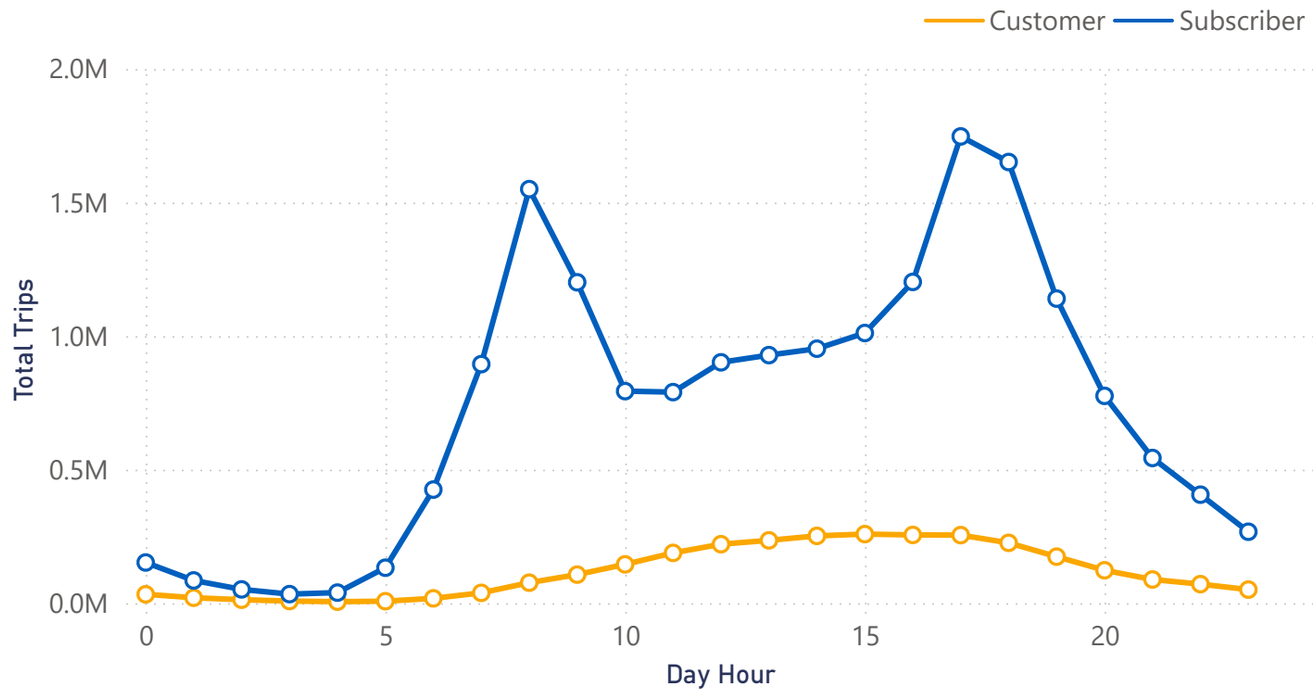
## Weekday

- [ ] False
- [ ] True

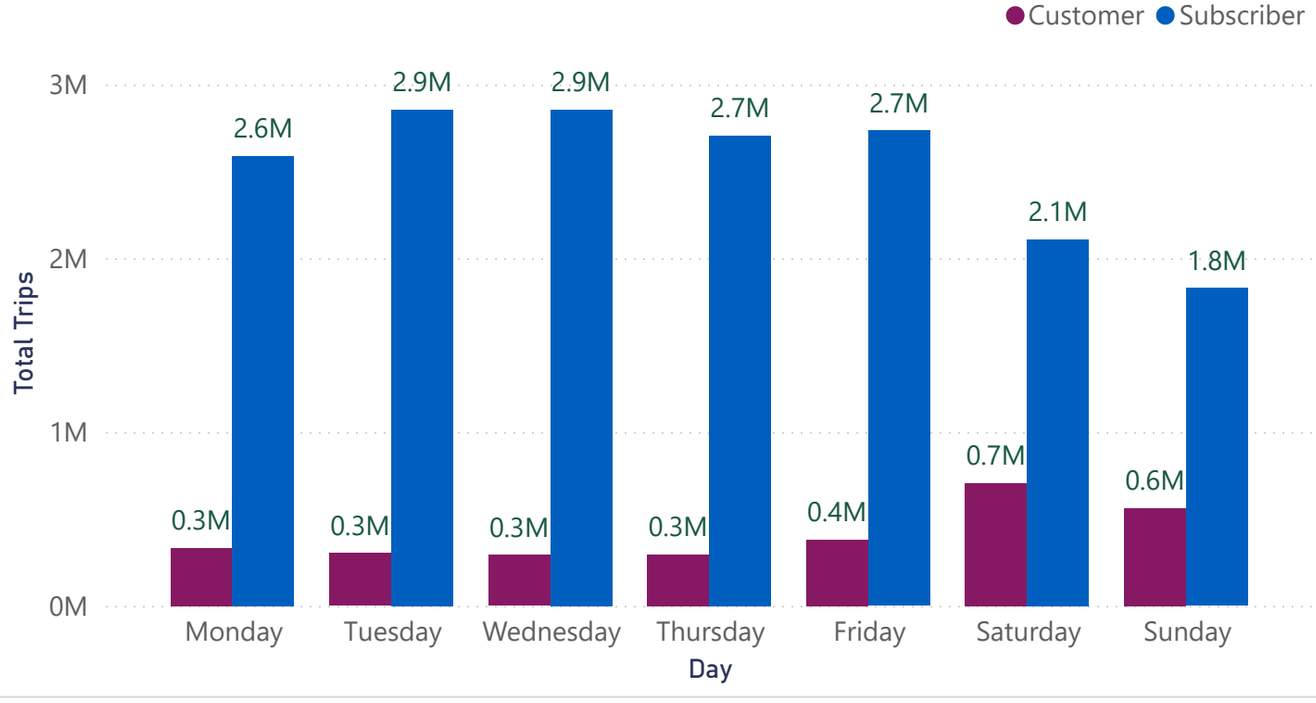
## Month

- [ ] January
- [ ] February
- [ ] March
- [ ] April
- [ ] May
- [ ] June
- [ ] July
- [ ] August
- [ ] September
- [ ] October
- [ ] November
- [ ] December

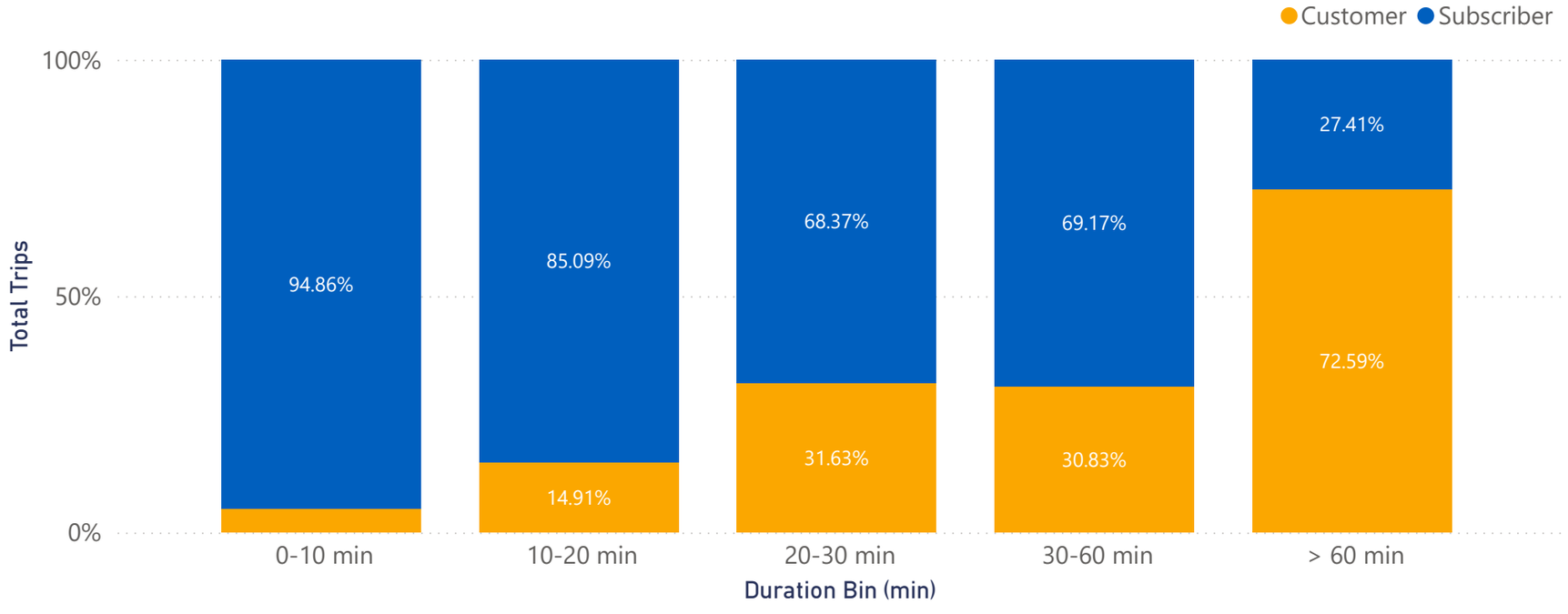
## Daily Trip Volume by User Type



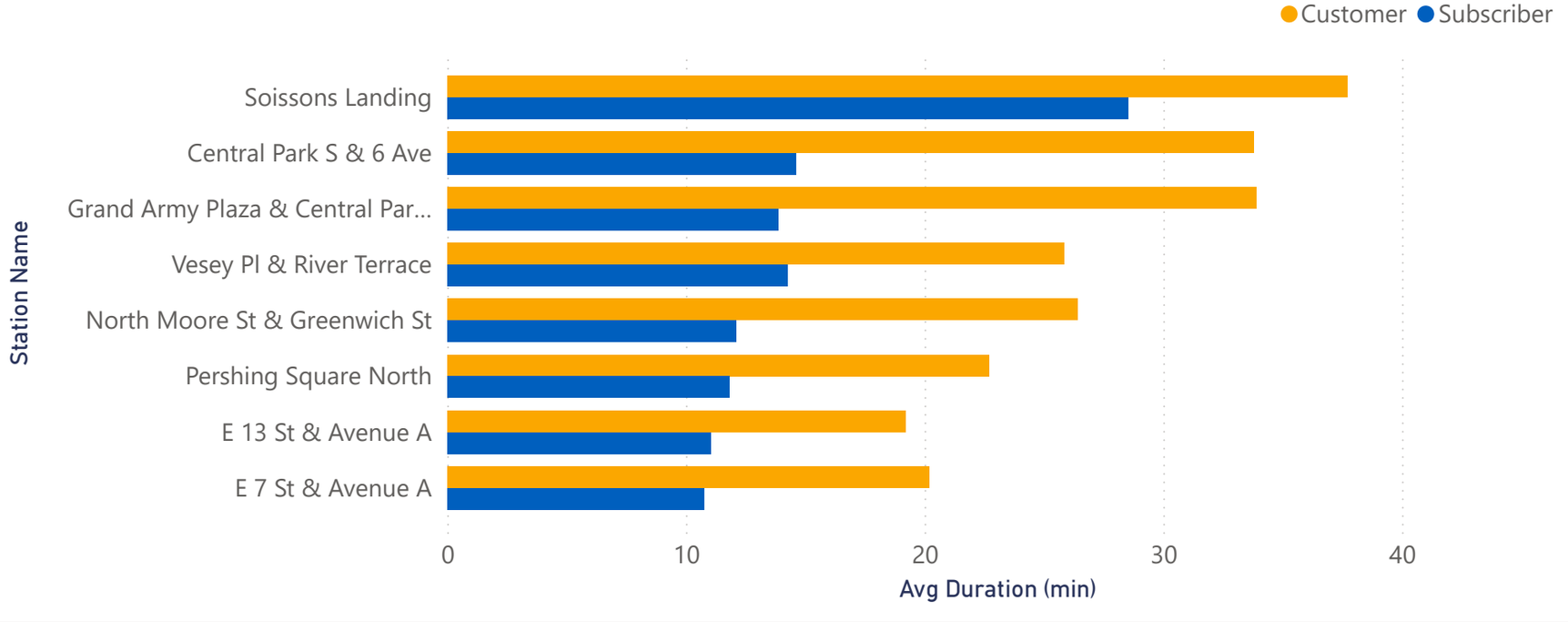

## Weekly Trip Volume by User Type

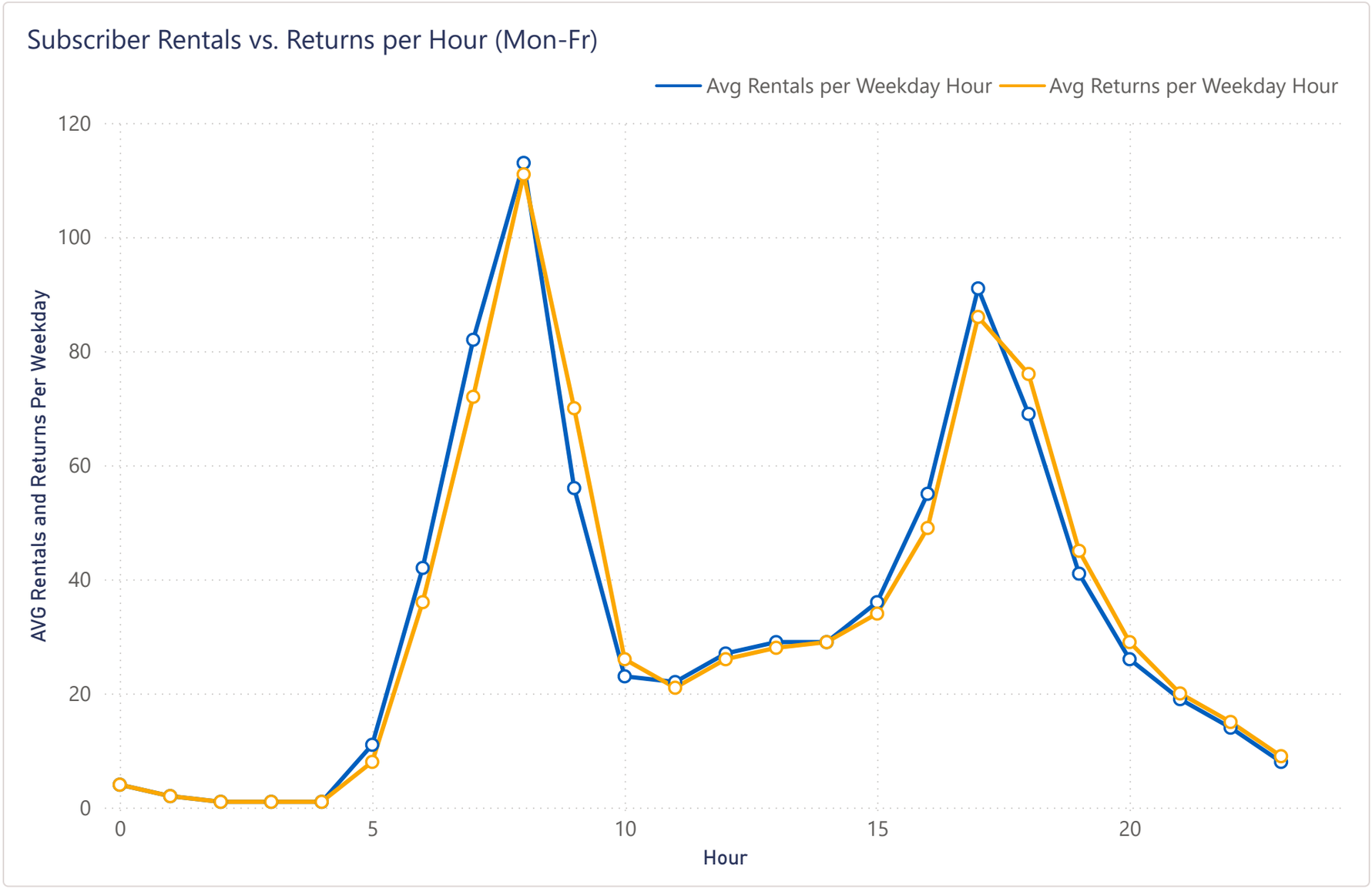


## Trip Duration Distribution by User Type (Whole Dataset)




## Average Trip Duration: Top Frequent Stations




Subscriber Rentals vs. Returns per Hour (Mon-Fr)
Avg Rentals per Weekday Hour
Avg Returns per Weekday Hour
AVG Rentals and Returns Per Weekday
120
100
80
60
40
20
0
0
5
10
15
20
Hour

# Strategic Insights & Recommendations

## Q2: Usage Patterns (Subscribers vs. Customers)

Key Findings:

- Subscribers (Commuters): Exhibit a classic "M-shaped" hourly curve (peaks at 8 AM & 5 PM) and are heavily skewed toward weekdays.
- Customers (Tourists): Show a flatter midday usage curve and a distinct 75% increase in volume on weekends.
- Action: Rebalance fleet from residential to business zones during rush hours; maintain tourist hub availability on weekends.

## Q4: Trip Duration Analysis

Key Findings:

- Overall Pattern: Subscribers' trips are heavily concentrated in the 0-60 minutes, reflecting quick, point-to-point commutes. Customers show a much wider distribution, with a significantly higher proportion of trips lasting 30–60+ minutes.
- Station-Level Differences: At commuter-heavy stations (e.g., E 7 St & Avenue A), the average duration for both user types is relatively low and similar. However, at tourist-heavy stations (e.g., Central Park South), the average duration for Customers spikes dramatically, while Subscribers maintain their short, efficient ride times.
- Action: Implement time-based pricing or incentives. For example, offer a "first 30 minutes free" perk to reinforce the Subscriber value proposition, while introducing premium "hourly passes" for Customers who frequently exceed the standard 30-minute window.

## Q5: Commute Flow Analysis (Subscribers, Mon-Fri)

Key Findings:

- The Morning Rush: Rentals (Starts) peak sharply at 8:00 AM, with Returns (Ends) following closely at 9:00 AM. This 1-hour lag represents the average morning commute duration.
- The Evening Rush: The pattern perfectly inverts in the evening. Rentals peak at 5:00 PM (leaving business districts), and Returns peak at 6:00 PM (arriving at residential areas).
- Midday Dormancy: Both metrics drop to their lowest points between 10:00 AM and 4:00 PM, confirming that Subscribers strictly use the system for commuting, not midday leisure.
- Actionable Recommendation: Optimize fleet redistribution to match this exact tidal flow. Ensure docks are empty at residential stations by 7:30 AM to accept morning returns, and full at business district stations by 4:30 PM to accommodate the evening exodus.